CLF-C02 · CONCEPTS BEFORE ANSWERS

Understand the topic. Choose with a reason.

A plain-English guide to cloud foundations, security, compute, storage, networking, data, operations, billing and migration—with examples that prepare you for unfamiliar questions.

13 sections · Companion to your 8 mock tests
Sources checked 16 September 2026

READ → EXPLAIN → PRACTICE

Use this alongside your eight tests.

Read one topic, then explain its key distinction aloud. Close the guide before answering practice questions. Open it again only to understand a mistake or a guess. You do not need to memorize service catalogs or implementation commands for this level.

24%
Cloud Concepts

30%
Security and Compliance

34%
Cloud Technology and Services

12%
Billing, Pricing and Support

AWS reference: CLF-C02 exam guide

Before this test	Read these sections
1 · Cloud foundations	1 · Foundations; 2 · Well-Architected; 12 · Adoption and migration.
2 · Security and responsibility	3 · Responsibility and IAM; 4 · Security tools; the monitoring comparison in 10.
3 · Compute and storage	5 · Compute; 6 · Storage; the deployment tools in 10.
4 · Networking, data and integration	7 · Networking; 8 · Data; 9 · Messaging and AI; migration tools in 12.
5 · Billing, pricing and support	11 · Billing; cloud economics in 1; Health and Trusted Advisor in 10.
6 · Mixed checkpoint; 7–8 · Full mocks	Revisit your uncertain topics, then use 13 · Final revision.

This is an independent learning companion, with original explanations and worked examples. It explains concepts without listing your mock-test answer keys. The topic mapping does not assume you have completed any test.

01 CORE CONCEPTS · MAINLY TEST 1

Cloud foundations

Cloud computing means obtaining computing resources over a network when you need them. Instead of buying every server in advance, a company can provision resources, use them, and release them. The useful question is: **what business problem does that flexibility solve?**

A small online shop might need little capacity most days and much more during a promotion. Buying hardware for the busiest day leaves capacity idle afterward. Cloud resources can reduce that mismatch. They also let teams experiment faster, reach other geographic markets, and spend less effort operating physical infrastructure.

AWS reference: Cloud benefits

Concept	Plain-English meaning	Clue in a question
Agility	How quickly you can try an idea or deliver a change.	Create an experiment in minutes.
Scalability	The ability to handle more work by increasing capacity.	Support growth from hundreds to thousands of users.
Elasticity	Adjust capacity up and down as demand changes.	Add capacity during a rush; remove it afterward.
Horizontal scaling	Add or remove machines or instances.	Two application servers become six.
Vertical scaling	Give a machine more or less CPU or memory.	Replace a small instance with a larger one.
High availability	Keep a service accessible despite component failures.	Run across separate Availability Zones.
Fault tolerance	Continue operating through a failure with minimal disruption.	Redundant components sustain the service.
Durability	Keep stored data from being lost.	Preserve the object even if hardware fails.

Availability and durability measure different things. A document can remain safely stored while temporarily inaccessible. Conversely, a system can be reachable while an application has accidentally deleted an important record. Backups, redundancy and recovery address different risks.

Cloud economics

Fixed versus variable expense: purchasing a data center creates substantial upfront cost; consumption pricing ties more expense to usage. **Economies of scale:** a provider can spread infrastructure costs across many customers. **Total cost of ownership (TCO):** compare the whole operating cost, including equipment, facilities, power, licensing and staff effort.

Rightsizing means matching resource size to the actual workload. Cloud spending still needs management: an oversized resource left running can waste money. The cheapest hourly price is not always the lowest total operating cost.

Service models and deployment models

Model	What you manage
IaaS: infrastructure as a service	With virtual machines such as EC2, you manage the guest operating system and application.
PaaS: platform as a service	Focus on the application while a provider manages more of the platform. Elastic Beanstalk is a useful example.
SaaS: software as a service	Use a finished application. You still manage appropriate user access, settings and data use.
Cloud / on-premises / hybrid	Cloud runs with a provider; on-premises runs in your own environment; hybrid connects both.

AWS reference: Cloud service and deployment models · Reliability pillar

> **Worked example · Growing demand versus failure protection**
>
> Adding instances during a promotion demonstrates elasticity. Spreading instances across Availability Zones helps availability. A scenario can involve both; choose the concept that answers the stated objective.

Before moving on: Explain scalability, elasticity and availability without using the same definition for all three.

02 DESIGN PRINCIPLES · TESTS 1, 6, 7 AND 8

The Well-Architected Framework

The AWS Well-Architected Framework is a set of questions and practices for evaluating a workload. A **workload** is an application or collection of components that delivers business value. The framework helps a team identify risks and make design decisions.

It has **six pillars**. Learn the goal of each, then connect a proposed action to that goal. A service can support several pillars, so a service name alone rarely identifies the answer.

AWS reference: Well-Architected pillars

Pillar	Question it asks	Concrete example
Operational excellence	How do we operate, learn and improve?	Automate deployments; update a runbook after an incident.
Security	How do we protect access, systems and data?	Use least privilege, encryption and layered controls.
Reliability	How do we keep functioning and recover?	Use redundancy, test recovery and replace failed resources.
Performance efficiency	How do we meet performance needs efficiently?	Choose an appropriate instance type, cache or database.
Cost optimization	How do we deliver the required value for less cost?	Remove idle resources and choose suitable purchasing options.
Sustainability	How do we reduce environmental impact?	Reduce unnecessary processing, storage and idle capacity.

AWS reference: Operational excellence · Security pillar · Reliability pillar · Performance efficiency · Cost optimization · Sustainability

How to distinguish the close answers

Performance efficiency versus cost optimization: choosing hardware to satisfy response-time requirements emphasizes performance; reducing the bill while meeting those requirements emphasizes cost. **Operational excellence versus reliability:** improving incident procedures emphasizes operations; recovering the service after a failure emphasizes reliability.

Cost optimization versus sustainability: the same action can improve both. If a question highlights spending, focus on cost. If it highlights energy use or environmental impact, focus on sustainability.

> **Worked example · One action, several benefits**
>
> A team turns off unused test servers overnight. Asked about reducing the AWS bill, choose cost optimization. Asked about reducing wasted compute and environmental impact, choose sustainability. Read the objective before matching a pillar.

Recognize these design ideas

Automate repeatable work: a repeatable process reduces manual variation. **Design for failure:** assume a component can fail and prepare recovery. **Loose coupling:** components communicate without needing to run in lockstep. **Measure and improve:** use evidence to revisit decisions as requirements change.

The **AWS Well-Architected Tool** supports workload reviews against the framework. It does not automatically guarantee that an application is secure or reliable.

AWS reference: Management and governance

Before moving on: Name the six pillars and give one original example for each. If two pillars seem plausible, explain which objective makes one the better answer.

03 SECURITY FUNDAMENTALS · MAINLY TEST 2

Shared responsibility and IAM

AWS secures the infrastructure it operates. You secure how you use it. The boundary moves as you choose services that AWS manages more fully.

A useful phrase is **AWS: security of the cloud; customer: security in the cloud.** AWS handles physical facilities and underlying infrastructure. Customers remain responsible for their data, access decisions and application configuration. AWS compliance certifications do not automatically make every customer application compliant.

Service	AWS handles	Customer still handles
EC2	Physical host, virtualization and underlying infrastructure.	Guest OS patches, application software, data, IAM and security-group rules.
Standard RDS managed database	Underlying infrastructure and managed database platform maintenance.	Data, database users, access, settings and appropriate maintenance choices.
Lambda	Compute infrastructure and service operation.	Function code, permissions, dependencies and supported runtime choices.
S3	Storage infrastructure and underlying platform.	Data classification, permissions, retention and suitable protection settings.

AWS reference: Shared responsibility · RDS security responsibilities · Lambda runtime responsibilities

Authentication and authorization

Authentication verifies identity: "Who are you?" **Authorization** determines permission: "What may you do?" Signing in successfully does not mean you may delete a database.

Least privilege means granting only the actions and resources needed for the job. **MFA**, or multi-factor authentication, adds another verification factor. Protect the account root user with MFA and use it only when a task requires it; use appropriate roles or workforce access for daily work.

IAM term	What it means
User	An identity in an AWS account; it may have long-term credentials.
Group	A collection of IAM users that can receive common permissions.
Role	An identity that an authorized person, application or service assumes to obtain temporary credentials.
Policy	A permissions document defining allowed or denied actions, resources and conditions.
Federation	Use an existing identity provider to access another system without duplicating everyone's credentials.

IAM term	What it means
IAM Identity Center	Central workforce access to AWS accounts and applications.
Amazon Cognito	Identity and sign-in capabilities for customers of your web or mobile application.

AWS reference: IAM best practices · Security services

How permissions questions work

For an ordinary IAM access question, begin with **default deny**: permission needs a relevant allow. An applicable **explicit deny overrides an allow**. Different policy types combine in different ways, so check whose identity, which resource and which action the question describes.

A **service control policy (SCP)** sets permission limits for accounts in AWS Organizations. It does not grant permissions by itself. An IAM allow cannot overcome an applicable SCP restriction.

AWS reference: IAM policy evaluation

> **Worked example · An application needs S3 access**
>
> An application on EC2 must read a particular bucket. Give its instance an IAM role with the necessary S3 permissions. The application can use temporary credentials; embedding an administrator's permanent access key would grant excessive access and create a credential-management problem.

Before moving on: Explain why "AWS manages this service" does not mean "AWS decides who may access my data."

04 THREATS, PROTECTION AND COMPLIANCE · TESTS 2 AND 6–8

Security tools: choose by the job

A security question usually names a problem: a threat, a vulnerable package, sensitive data, excessive access or a need for compliance evidence. These are different jobs.

Need	Best service to recognize
Detect suspicious or malicious activity	**GuardDuty**: threat detection.
Find software vulnerabilities in supported workloads	**Inspector**: vulnerability assessment.
Discover sensitive information in S3	**Macie**: sensitive-data discovery.
Consolidate findings and evaluate security posture	**Security Hub / Security Hub CSPM**.
Filter malicious web requests	**AWS WAF**: web application firewall rules.
Protect against distributed denial-of-service attacks	**AWS Shield**. Shield Standard provides baseline protection.
Manage cryptographic encryption keys	**AWS KMS**: Key Management Service.
Store and rotate application passwords or secrets	**Secrets Manager**.
Manage TLS certificates	**Certificate Manager (ACM)**.
Obtain AWS compliance reports and agreements	**AWS Artifact**.
Manage security policies across accounts	**AWS Firewall Manager**.

AWS reference: Security services · GuardDuty · Inspector

IAM Access Analyzer is the clue for reviewing resource access beyond a trust boundary, such as another account or the public. It also validates policies and can generate policies based on observed access activity. CloudTrail supplies activity records; Access Analyzer uses that evidence to help refine permissions.

AWS reference: IAM Access Analyzer

Encryption in simple terms

At rest means data while stored. **In transit** means data while moving between systems, commonly protected with TLS. Encryption and access control solve different problems: data can be encrypted yet still accessible to someone who has permission to decrypt it.

AWS reference: Security pillar

> **Worked example · Three different findings**
>
> An outdated software package suggests Inspector. Unusual account activity suggests GuardDuty. Sensitive personal information in an S3 bucket suggests Macie. All concern security, but the required analysis determines the service.

Before moving on: Without looking back, distinguish KMS, Secrets Manager and ACM in one sentence each.

05 RUNNING APPLICATIONS · MAINLY TEST 3

Compute and containers

Compute is the processing capacity that runs your application. Start with the amount of control you need and the operational work you want AWS to manage.

Service	Use it when the question asks for...
Amazon EC2	A virtual server with control over its operating system and installed software.
AWS Lambda	Event-driven function execution without managing servers.
Amazon Lightsail	A simplified bundle for a small website or application.
Elastic Beanstalk	Deploy application code while AWS helps handle provisioning, scaling and application hosting.
EC2 Auto Scaling	Adjust the number of EC2 instances and replace unhealthy ones.
Elastic Load Balancing	Distribute incoming traffic among healthy targets.

AWS reference: Compute services

Serverless means you do not provision and administer the underlying servers. Servers still exist; you still manage your code, permissions and service configuration. An always-running workload needing a custom operating system points toward EC2; a short task triggered by an event often points toward Lambda.

Container vocabulary

A **container image** packages application code and dependencies. A **registry** stores images. An **orchestrator** schedules and manages running containers. The **compute engine** supplies the capacity on which they run.

Service	Remember the role
Amazon ECR	Registry: stores container images.
Amazon ECS	AWS container orchestration.
Amazon EKS	Managed Kubernetes orchestration.
AWS Fargate	Serverless compute for supported ECS and EKS container workloads.

AWS reference: Container services

EKS and Fargate answer different questions. "We must use Kubernetes" identifies EKS. "We do not want to manage the container hosts" identifies Fargate where supported. "We need somewhere to store the image" identifies ECR.

Instance families

General purpose balances resources. **Compute optimized** emphasizes CPU. **Memory optimized** emphasizes RAM. **Storage optimized** fits demanding local storage access. **Accelerated computing** uses specialized processors such as GPUs. Match the resource bottleneck; do not memorize every model number.

AWS reference: Compute services

> **Worked example · Traffic suddenly increases**
>
> The load balancer spreads requests across available servers. Auto Scaling adds servers when its configured conditions are met. A load balancer alone does not create extra processing capacity.

Before moving on: Explain EC2, Lambda, ECR, ECS, EKS and Fargate as six different jobs.

06 PERSISTENCE AND RECOVERY · MAINLY TEST 3

Storage: object, block and file

Ask how the application expects to access data. **Objects** are retrieved by identifiers; **blocks** behave like a disk; **files** live in a shared filesystem with directories.

Service	Storage type	Typical requirement
S3	Object	Images, documents, backups and data-lake objects accessed through APIs.
EBS	Persistent block	A disk volume for an EC2 instance. The volume is associated with an Availability Zone.
EC2 instance store	Temporary local block	Fast scratch space or a cache that can be recreated.
EFS	Shared file	A managed NFS filesystem for multiple Linux-oriented workloads.
FSx for Windows File Server	Shared file	Windows SMB file shares and integration with Active Directory.
AWS Backup	Backup management	Central policies and backup management across supported services.

AWS reference: Storage services · Instance store

Persistence matters: EBS data survives an instance stop, although volume deletion on instance termination depends on settings. Instance-store data can be lost when an instance stops, terminates or its underlying storage fails. A normal reboot is a different event.

An EBS snapshot is a point-in-time backup. A backup provides a recovery copy; it does not by itself keep a running application continuously available.

S3 storage classes: read the access requirement

Class	Useful exam distinction
S3 Standard	Frequently accessed objects.
S3 Intelligent-Tiering	Access patterns change or are difficult to predict; eligible objects move between access tiers.
S3 Standard-IA	Infrequent access with immediate retrieval; consider retrieval charges.
S3 One Zone-IA	Infrequent access in one AZ; consider whether data can be recreated if that AZ is lost.
S3 Glacier Instant Retrieval	Rarely accessed archive data that still needs millisecond retrieval.

Class	Useful exam distinction
S3 Glacier Flexible Retrieval	Archive data that can tolerate a restore process; retrieval options vary.
S3 Glacier Deep Archive	Very infrequent access where waiting hours for restoration is acceptable.

AWS reference: S3 storage classes

Lifecycle rules automate transitions or expiration according to a policy. **Versioning** preserves versions of an object and can help recover from accidental changes. Neither is a substitute for designing the permissions and retention rules correctly.

Storage cost can include more than stored gigabytes: requests, retrieval, minimum retention periods, transitions and transfer can matter. “Archive” does not always mean “slow”: Glacier Instant Retrieval exists.

> **Worked example · Shared files or uploaded documents?**
>
> Several Linux servers must see the same shared directory: EFS fits. A website stores uploaded photographs and retrieves them by object key: S3 fits. A single EC2 server needs a disk for its operating system: EBS fits.

Before moving on: Name the storage type before choosing a service. Then check persistence, sharing and retrieval time.

07 LOCATION, CONNECTIVITY AND TRAFFIC · MAINLY TEST 4

Networking and global infrastructure

Networking questions usually ask where resources run, how traffic reaches them, or how to restrict that traffic. Keep these three decisions separate.

Term	Meaning and consequence
Region	A geographic area containing multiple Availability Zones. Choose based on latency, service availability, cost and data-residency requirements.
Availability Zone (AZ)	One or more discrete data centers with independent infrastructure. Using multiple AZs helps limit the impact of a local failure.
Edge location	A location closer to end users that can support edge services such as CloudFront content delivery.
VPC	A logically isolated virtual network within a Region.
Subnet	A range of addresses within a VPC; each subnet belongs to one AZ.

AWS reference: CLF-C02 exam guide · Networking services

Multi-AZ and multi-Region are different choices. Multiple AZs within one Region commonly support high availability. Multiple Regions can address broader disaster recovery, geographic reach or other requirements, with additional design work and cost.

Services and connections

Requirement	Service or component
Resolve a domain name and select where DNS points	**Route 53**: DNS, routing policies and health checks.
Deliver cached web content nearer users	**CloudFront**: content delivery network (CDN).
Improve paths to application endpoints over the AWS global network	**Global Accelerator**: network routing, not a content cache.
Public IPv4 connectivity for a VPC	**Internet gateway**, plus appropriate routes, addressing and security rules.
Private-subnet IPv4 resources initiate internet connections	**Public NAT gateway**, with the required public-subnet and internet-gateway routing.

Requirement	Service or component
Private access to supported AWS services	**VPC endpoints**, avoiding a public internet path.
Encrypted site-to-site tunnel over the internet	**AWS Site-to-Site VPN**.
Dedicated private connection from a location to AWS	**AWS Direct Connect**. Encryption is not automatic merely because the connection is private.

AWS reference: Networking services · VPC security

A **public subnet** has a direct route to an internet gateway. This does not make every resource publicly reachable: the resource also needs suitable addressing, routing and permissions. A NAT gateway lets private resources initiate supported outbound connections; it is not a general way for internet users to initiate connections to those instances.

Security groups versus network ACLs

Feature	Security group	Network ACL
Scope	Attached to supported resources / network interfaces.	Applied at the subnet boundary.
Rules	Allow rules.	Allow and deny rules.
Connection tracking	Stateful: response traffic for an allowed connection is tracked.	Stateless: return traffic needs appropriate rules too.

AWS reference: Security groups · Network ACLs

> **Worked example · A website is slow for overseas readers**
>
> If repeatedly requested images need to reach users faster, CloudFront's caching is the relevant clue. Route 53 resolves and routes DNS, but DNS alone is not a content cache. If the question instead asks about a private office-to-AWS connection, investigate Direct Connect.

Before moving on: Explain Region, AZ and edge location, then distinguish DNS, caching and private connectivity.

08 OPERATIONAL DATA VERSUS ANALYSIS · TESTS 4, 6, 7 AND 8

Databases and analytics

A database serves application data. Analytics examines data to answer broader questions. First decide whether the question describes a live application, a cache, a reporting system or a query over files.

Service	What to recognize
RDS	Managed relational databases: tables, SQL and relational application workloads.
Aurora	AWS relational database technology compatible with MySQL and PostgreSQL.
DynamoDB	Managed serverless NoSQL database for key-value and document access patterns.
ElastiCache	In-memory caching to reduce repeated database work and improve latency.
Neptune	Graph data: relationships among people, accounts, devices or other entities.
Redshift	Data warehousing and analytical queries over substantial datasets.

AWS reference: Database services

For an application maintaining relational records, RDS or Aurora is a natural starting point. For a key-based lookup workload designed for NoSQL, recognize DynamoDB. For frequently reused query results, a cache can help. A graph problem emphasizes relationships and traversals.

The RDS distinction that often causes mistakes

A classic RDS Multi-AZ DB instance deployment has a standby for failover; that standby does not serve application reads. **Read replicas** are commonly the clue for read scaling. However, **RDS Multi-AZ DB clusters** are a different deployment type whose reader instances can also serve reads. Avoid the absolute statement "Multi-AZ never serves reads."

AWS reference: RDS Multi-AZ deployments

> **Worked example · Availability or reporting capacity?**
>
> A relational database must recover from an AZ failure: look for an appropriate Multi-AZ deployment. A primary database is overloaded by read-only reporting: read replicas may fit. Identify the problem before choosing the feature.

Analytics services

Need	Service
Query data in S3 using SQL without managing query servers	**Athena**.
Discover, catalog, clean and transform data	**AWS Glue**; ETL means extract, transform and load.
Run frameworks such as Apache Spark or Hadoop	**Amazon EMR**.
Ingest and process streams of events	**Amazon Kinesis Data Streams**.
Search and analyze indexed data or logs	**Amazon OpenSearch Service**.
Build business dashboards and visual reports	**Amazon QuickSight** (written as Quick Sight in the exam guide).

AWS reference: Analytics services

A common analytical flow is: data arrives, is stored, is cataloged or transformed, is queried, and is visualized. Glue, Athena and dashboard tools can therefore work together. They do not all perform the same step.

> **Before moving on:** Distinguish RDS, DynamoDB, Redshift, Athena and ElastiCache using one requirement for each.

09 RECOGNIZE THE PURPOSE · TESTS 4 AND 6–8

Messaging, AI and business services

Integration services connect components. A queue, a notification, an event and a workflow represent different ways of coordinating work.

Service	Main job	Useful clue
SQS	Queue messages until a consumer processes them.	Buffer work; absorb a surge; decouple workers.
SNS	Publish notifications to subscribers.	Send one notification to multiple subscribers.
EventBridge	Route events according to matching rules.	React to an event from an application, AWS service or supported SaaS source.
Step Functions	Coordinate a multi-step workflow.	Sequence tasks, branch, track state and handle retries.
API Gateway	Publish and manage application APIs.	Provide a managed API entry point.

AWS reference: Application integration · Networking services

> **Worked example · Process orders without losing the backlog**
>
> Put order-processing work in SQS so workers can process queued messages at their own pace. If several independent systems should receive an order notification, SNS can fan it out. If a process requires several coordinated steps, Step Functions can manage that workflow.

Decoupling means one component can work without requiring another component to finish immediately. This can make it easier to scale parts independently and handle temporary unavailability.

AI: connect the input to the output

Input or task	Service to recognize
Scanned documents → extracted text, forms or tables	**Textract**.
Written text → sentiment, entities or language insights	**Comprehend**.

Input or task	Service to recognize
Speech → written transcript	**Transcribe**.
Written text → spoken audio	**Polly**.
Text in one language → another language	**Translate**.
Images or video → detected visual content	**Rekognition**.
Conversational voice or text interface	**Lex**.
Build, train and deploy machine-learning models	**SageMaker AI**.
Build generative-AI applications using foundation models	**Bedrock**.

AWS reference: AI and machine learning services

Business-service recognition: Amazon Connect supports contact-center experiences; Amazon SES sends application email; Amazon WorkSpaces provides virtual desktops. These are different from running a general-purpose server on EC2.

AWS reference: CLF-C02 exam guide

Before moving on: If a question asks to extract fields from an invoice, explain why Textract fits better than a translation or sentiment-analysis service.

10 OBSERVE, AUDIT, CONFIGURE AND RELEASE · TESTS 2, 3 AND 6–8

Monitoring, governance and deployment

Several AWS names begin with “Cloud.” Translate the requirement into a verb: **monitor, audit, evaluate configuration, or provision.**

Service	Job	Example question
CloudWatch	Metrics, logs, dashboards and alarms.	Is CPU high? Has application latency crossed a threshold?
CloudTrail	Record account activity and API events.	Which identity performed this action, and when?
AWS Config	Track resource configuration and evaluate configured rules.	How was this resource configured? Does it meet the rule?
CloudFormation	Provision infrastructure from templates.	How can we recreate this environment consistently?
AWS Health	AWS service events and relevant resource or account impacts.	Is AWS maintenance or an AWS service event affecting us?
Trusted Advisor	Recommendations and checks for AWS usage.	Where might this environment improve cost or other practices?

AWS reference: CloudWatch · CloudTrail · AWS Config · CloudFormation · Management and governance

> **Worked example · A resource setting changed**
>
> Use CloudTrail to investigate who made the API call. Use Config to examine the resource’s configuration and changes. Use CloudWatch to monitor resulting application behavior. These sources can complement one another.

Deployment tools and access methods

Tool or method	Purpose
Management Console	Interact with AWS through the browser interface.
AWS CLI	Call AWS operations using command-line commands.
SDK	A software library that helps application code call AWS APIs.
Infrastructure as code	Describe infrastructure in versionable code or templates; CloudFormation is an AWS example.

Tool or method	Purpose
CodeBuild	Compile code, run tests and produce build artifacts.
CodePipeline	Coordinate release stages such as source, build and deployment.
X-Ray	Trace requests through distributed application components.
Systems Manager	Manage operational tasks across supported nodes and resources, including automation and patching capabilities.

AWS reference: Developer tools · CLF-C02 exam guide

Provisioning versus configuration history: CloudFormation describes and creates resources; Config records and evaluates their configurations. **Monitoring versus tracing:** an alarm tells you a metric crossed a threshold; a trace helps follow a request across components.

Organizations manages multiple AWS accounts and supports consolidated billing and policies. **Control Tower** helps establish and govern a multi-account environment. Account separation can isolate workloads, permissions and cost ownership.

AWS reference: Management and governance

Before moving on: Finish these sentences: CloudWatch watches ___. CloudTrail records ___. Config evaluates ___. CloudFormation creates ___.

11 MATCH THE COST TOOL TO THE QUESTION · MAINLY TEST 5

Billing, pricing and support

Separate three questions: **How will we buy capacity? How will we understand spending? Where will we get help?** A discount, a budget and a support plan solve different problems.

Compute purchasing choices

Option	When it fits	Important limit
On-Demand	Short-lived or uncertain usage; no long-term usage commitment.	You pay the applicable usage rate.
Spot	Interruptible work that can restart or checkpoint.	Spare capacity can be interrupted; plan for it.
Reserved Instances (RIs)	Eligible predictable instance usage in exchange for a term commitment.	An RI is primarily a billing benefit; capacity benefits depend on scope and type.
Compute Savings Plans	Commit to eligible compute spending per hour for one or three years.	The commitment remains payable even if eligible usage drops.
On-Demand Capacity Reservation	Need matching EC2 capacity in a particular AZ.	Capacity assurance is not automatically a discount.
Dedicated Hosts	Need a dedicated physical host and host-level placement or licensing visibility.	Different from simply choosing a larger instance.
Dedicated Instances	Need instances on hardware dedicated to your account.	You do not get the same host-level control as a Dedicated Host.

AWS reference: EC2 purchasing options

Compute Savings Plans can apply across eligible EC2, Fargate and Lambda usage. **EC2 Instance Savings Plans** are tied to an instance family in a Region and offer less flexibility. Do not assume every Savings Plan covers every AWS charge.

Reserved capacity and discounted pricing are separate ideas. A Regional EC2 RI provides a billing discount without reserving capacity. A matching zonal EC2 RI can also provide capacity reservation. Always read whether the scenario asks for lower cost or guaranteed capacity.

AWS reference: Savings Plans types · EC2 purchasing options

> **Worked example · Three workloads, three purchasing answers**
>
> An experiment has unpredictable usage: On-Demand may fit. A batch job can pause and retry: Spot may fit. A stable baseline will run for a long time: evaluate an appropriate commitment discount. Choose based on the workload constraints, not the largest advertised discount.

Cost-management tools

Question	Tool
What might this proposed setup cost?	**AWS Pricing Calculator**: estimate a planned configuration.
What did we spend, and what drove the change?	**Cost Explorer**: analyze usage, trends, filtering and forecasts.
Can I be notified when actual or forecast spending crosses a threshold?	**AWS Budgets**.
Where can I obtain detailed billing records for analysis?	**AWS Data Exports / Cost and Usage Report (CUR)**; CUR 2.0 is the recommended detailed export.
How can costs be grouped by team or project?	**Activated cost allocation tags** and suitable account organization.
How can multiple accounts share a consolidated bill?	**AWS Organizations consolidated billing**.

AWS reference: Exam guide: billing and support · Cost Explorer · AWS Budgets · Cost and usage exports

A budget is not an automatic hard spending cap. Alerts require configuration and billing data is not instantaneous. Budget actions can perform configured actions, such as applying certain policies or stopping supported resources, but simply creating a budget does not guarantee that every service stops at a dollar amount.

Cost Explorer investigates actual usage and can forecast future cost; Pricing Calculator estimates a configuration you describe. A forecast and a proposed architecture estimate are not interchangeable.

AWS reference: AWS Budgets · Cost Explorer

Charges people overlook

Stopping an EC2 instance generally stops its On-Demand instance-compute charge, but associated storage and other resources can continue to cost money. An RI or Savings Plan commitment is not canceled by stopping an instance. S3 charges can include requests and retrieval. Data-transfer charges depend on the route and service; do not assume all transfers within AWS are free.

Bring Your Own License (BYOL) uses eligible existing licenses under their terms. **License included** includes applicable software licensing in the service price. Check the scenario's licensing constraint before selecting a hosting option.

AWS reference: Exam guide: billing and support · EC2 purchasing options · S3 storage classes · EC2 lifecycle and billing

Support: understand the capabilities

Need	What to recognize
Account and billing help, documentation and community resources	Basic Support provides baseline support resources; it does not include general technical support cases.
Technical troubleshooting by AWS Support	An appropriate paid support plan.
A designated Technical Account Manager (TAM)	Enterprise-level support; current Enterprise Support includes a designated TAM.
Community questions and technical knowledge	AWS re:Post and AWS documentation.
Third-party products or specialist delivery help	AWS Marketplace and AWS Partners serve different purchasing and expertise needs.

> **Checked 16 September 2026:** the current comparison lists Basic, Business Support+, Enterprise Support and Unified Operations. The CLF-C02 guide still includes older examples such as Developer, Business and Enterprise On-Ramp. Recognize the capabilities and the wording in the question; use the current comparison when checking today's offerings.

AWS reference: Current AWS Support plans · Exam guide: billing and support

Before moving on: Choose one tool for estimating, one for investigating costs, one for threshold alerts and one for detailed billing exports.

12 ORGANIZATIONAL READINESS AND MOVING WORKLOADS · TESTS 1, 4 AND 6–8

Cloud adoption and migration

Cloud adoption changes people and operating processes as well as technology. Migration questions ask what will change, what will stay the same and what needs to move.

Cloud Adoption Framework (CAF)

CAF evaluates organizational capabilities for cloud transformation. Its six **perspectives** are different from the six **Well-Architected pillars**, which assess workloads.

CAF perspective	Focus
Business	Connect cloud investments to business outcomes.
People	Skills, culture, leadership and organizational change.
Governance	Coordinate priorities, investment, risk and oversight.
Platform	Build and evolve the technical cloud platform.
Security	Protect data and workloads and manage security capabilities.
Operations	Deliver and operate cloud services to meet business needs.

AWS reference: CAF perspectives

The seven migration strategies

Strategy	What changes	Simple example
Rehost	Hosting location, with little application redesign.	Move an existing server-based application to EC2.
Replatform	Make selected platform improvements.	Move a self-managed database to RDS with limited application change.
Refactor / re-architect	Redesign significant parts of the application.	Break a monolith into independently operated services.
Repurchase	Replace the application product.	Move from an installed package to a suitable SaaS product.
Retain	Keep the workload where it is for now.	Delay migration because of dependencies.
Retire	Stop using an unnecessary workload.	Decommission an obsolete application.

Strategy	What changes	Simple example
Relocate	Move an existing environment with minimal architectural change.	Move workloads to a compatible platform environment.

AWS reference: The seven migration strategies

Match the migration tool to the material being moved

Service	Remember
AWS Database Migration Service (DMS)	Move and replicate database data; supported ongoing replication can reduce migration downtime.
Schema conversion tools, such as AWS SCT or DMS Schema Conversion	Assess or convert schema and code when changing database engines; manual work may still be needed.
AWS Application Migration Service (MGN), now AWS Transform MGN	Server/application rehosting.
AWS DataSync	Move file or object data between supported storage locations.
AWS Storage Gateway	Connect on-premises applications to AWS-backed storage for ongoing hybrid use.

AWS reference: Migration services · The seven migration strategies · Storage services

> **Worked example · Moving a database versus a server**
>
> If the requirement is to replicate database changes during migration, DMS is the relevant clue. If it is to move a server-based application largely as it is, look for application migration/rehosting. If the database engine changes, consider schema conversion as a separate requirement.

Before moving on: Explain rehost, replatform and refactor without saying only “move to the cloud.”

13 USE AFTER A TOPIC OR BEFORE A FRESH MOCK

Final revision: turn knowledge into answers

For each service, learn three things: **the problem it solves, a clue that identifies it, and the similar service you could confuse it with.**

Reason through a question in four steps

1. **Name the job.** Is the task to store, run, connect, monitor, audit, secure, migrate or estimate?
2. **Underline the constraint.** Examples: temporary credentials, shared files, no server management, immediate retrieval, or interruption tolerance.
3. **Eliminate the nearest wrong answer.** Explain the requirement it fails, even if it belongs to the correct service category.
4. **Check the requested number of answers.** A choose-two question needs two answers that independently satisfy the stated requirements.

Ten distinctions to recall without notes

Pair	Distinction to remember
Elasticity / high availability	Adapting capacity to demand / keeping the service accessible through failures.
Authentication / authorization	Verify identity / decide permissions.
IAM Identity Center / Cognito	Workforce access / customer application identities.
CloudWatch / CloudTrail / Config	Operational telemetry / API activity / resource configuration.
GuardDuty / Inspector / Macie	Threats / vulnerabilities / sensitive S3 data.
S3 / EBS / EFS	Objects / block disk / shared files.
ECR / ECS or EKS / Fargate	Store images / orchestrate containers / provide serverless container compute.
SQS / SNS / EventBridge	Queue work / notify subscribers / route events.
Multi-AZ / read replicas	Availability is the usual clue / read scaling is the usual clue; check the RDS deployment type.
Calculator / Explorer / Budgets	Estimate a setup / analyze cost / track thresholds and alerts.

After a practice test

Review both incorrect answers and correct guesses. For each, write one sentence: **"The requirement was ___; ___ fits because ___; ___ fails because ___."** Re-read only the relevant topic, then try fresh questions. Remembering an old answer letter is weaker evidence than explaining a new scenario.

The exam uses a scaled score; a practice percentage does not directly convert into an AWS score. Unanswered exam questions receive no credit, and there is no penalty for guessing. Answer every question after eliminating options where possible.

AWS reference: CLF-C02 exam guide

> **Your next step:** Read Cloud foundations and the Well-Architected Framework, then take Test 1 without notes. If you already understand those ideas, move to your least confident topic and its matching test.

Original study explanations based on the linked AWS references. Names, plans and service features can change; dated notes identify known naming differences. This file is a separate reading companion and does not change or reset your exam progress.